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Research paper

A collaborative exercise on DNA methylation-based age prediction and body fluid typing

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ABSTRACT

DNA methylation has become one of the most useful biomarkers for age prediction and body fluid identification in the forensic field. Therefore, several assays have been developed to detect age-associated and body fluid-specific DNA methylation changes. Among the many methods developed, SNaPshot-based assays should be particularly useful in forensic laboratories, as they permit multiplex analysis and use the same capillary electrophoresis instrumentation as STR analysis. However, technical validation of any developed assays is crucial for their proper integration into routine forensic workflow. In the present collaborative exercise, two SNaPshot multiplex assays for age prediction and a SNaPshot multiplex for body fluid identification were tested in twelve laboratories. The experimental set-up of the exercise was designed to reflect the entire workflow of SNaPshot-based methylation analysis and involved four increasingly complex tasks designed to detect potential factors influencing methylation measurements. The results of body fluid identification from each laboratory provided sufficient information to determine appropriate age prediction methods in subsequent analysis. In age prediction, systematic measurement differences resulting from the type of genetic analyzer used were identified as the biggest cause of DNA methylation variation between laboratories. Also, the use of a buffer that ensures a high ratio of specific to non-specific primer binding resulted in changes in DNA methylation measurement, especially when using degenerate primers in the PCR reaction. In addition, high input volumes of bisulfite-converted DNA often caused PCR failure, presumably due to carry-over of PCR inhibitors from the bisulfite conversion reaction. The proficiency of the analysts and experimental conditions for efficient SNaPshot reactions were also important for consistent DNA methylation measurement. Several bisulfite conversion kits were used for this study, but differences resulting from the use of any specific kit were not clearly discerned. Even when different

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experimental settings were used in each laboratory, a positive outcome of the study was a mean absolute age prediction error amongst participant's data of only 2.7 years for semen, 5.0 years for blood and 3.8 years for saliva.

1. Introduction

In the field of forensic genetics, age prediction is an important tool to narrow the range of potential suspects in a crime. Furthermore, the predicted minimum age may help to determine the criminal responsibility of a person and thus the legal framework to be applied [1]. Currently, DNA methylation is one of the most promising age predictive biomarkers [2]. A tissue-wide clock which enables age prediction across a broad spectrum of tissues and cells is available [3], but the need to generate DNA methylation profiles at hundreds of CpG sites using Beadchip array technology limits its application in forensic casework. Instead, tissue-specific age prediction models e.g., for blood, saliva, and semen, have been proposed, typically using small sets of loci containing age-associated CpG sites for a given body fluid [4–7].

To analyze a smaller number of CpG sites for these models, targeted bisulfite sequencing methods such as pyrosequencing [7–12], EpiTYPER [13,14], SNaPshot [6,15–17], methylation-sensitive high-resolution melt analysis [18] and massively parallel sequencing (MPS) [19–22] can be used. Among these methods, SNaPshot and MPS are often favored in forensic laboratories because of their increased multiplexing capability. However, MPS has the disadvantage of pooling multiple samples into a single reaction to reduce cost, with a very different workflow to STR analysis. Therefore, it may be longer to go from sample preparation to results, and extensive training of laboratory personnel in MPS processing may be necessary. In comparison, the SNaPshot system based on the dideoxy single-base extension (SBE) of an unlabeled oligonucleotide primer can perform multiplexed assays via capillary electrophoresis (CE), which is a standard method used in all forensic genetic laboratories. Therefore, the MPS method may be more useful in missing persons programs or mass disasters with a unique set of samples, and a single casework sample including an urgent criminal casework sample may be better suited to analysis with SNaPshot-based methods.

Several SNaPshot multiplex assays have been developed for age prediction from forensically relevant body fluids: i) a multiplex assay composed of five CpGs in *ELOVL2, FHL2, KLF14, MIR29B2CHG* (formerly known as *C1orf132*) and *TRIM59* genes was used for age prediction using blood [16]; ii) a multiplex assay composed of seven CpGs in *PTPN7, SST, CNGA3, KLF14, TSSK6, TBR1*, and *SLC12A5* genes was used for age prediction in saliva [17]; and iii) a multiplex assay composed of 3 CpGs including cg12837463, and two CpGs in *TTC7B* and *NOX4/FOLH1B* genes was used for age prediction from semen [6,23]. Age prediction models using these CpG markers all provided high prediction accuracy with a mean absolute error (MAE) of less than five years [6,16,17]. Interestingly, the 5-CpG assay for age prediction from blood can also be applied to the age prediction of saliva and buccal swabs with an accuracy of less than four years MAE [16].

It should be noted that in certain cases the body fluid type of unknown evidential samples may be difficult to determine using classical techniques. However, in the last few years, new methods have been developed to identify body fluids with improved sensitivity. These methods include cell type-specific RNA expression analysis [24–29], tissue-specific DNA methylation detection [30–35], protein profiling [36,37] and the detection of the microbiomes that reside on or within tissues [38–41]. However, to simplify the workflow, especially for a combined investigation of age and body fluid, it is preferable to use DNA methylation biomarkers, as both can be analyzed with the same technology.

In a collaborative exercise in 2016 [42], we evaluated the performance of a SNaPshot multiplex assay for differentiation of semen, blood, vaginal fluid, and saliva based on 7 CpG markers developed and published previously. Collaborative exercises provide participating laboratories with an opportunity to become acquainted with the analysis procedure, to identify challenges, and to suggest optimized interpretation guidelines [42]. Based on this knowledge, we have optimized the SNaPshot multiplex assay and improved it by adding two additional CpG markers for the identification of menstrual blood [43].

In 2019, a cooperative network of twelve forensic laboratories was established to undertake a collaborative project on DNA methylation-based age prediction and body fluid typing. The collaborative exercise described here, aimed to test the laboratory workflow, and comprised DNA extraction, DNA quantification, and DNA methylation-based body fluid identification steps followed by age prediction using the set of markers appropriate for the tissue. It was anticipated that this exercise could show the capabilities of the procedure when implemented in forensic laboratories. For body fluid identification, the improved SNaPshot multiplex assay described above was used with a minor modification [43]. Two SNaPshot multiplex assays were applied for age prediction, one for semen (3 CpGs) and the other for blood and saliva (5 CpGs). The participating laboratories received protocols, multiplex primer sets, standardized test samples, and were tasked with performing analyses according to the guidelines provided. The results were collected, anonymously evaluated and are presented here.

2. Materials and methods

2.1. Samples and materials provided

The SNaPshot workflow consists of five consecutive steps: DNA extraction, bisulfite conversion, multiplex PCR, multiplex SBE and CE. The exercise was divided into four sequential parts that were increasingly complex (Table 1). Within each part, the number of steps required by each laboratory increased. Part 1 involved only the CE of provided SBE products; part 2 involved additional multiplex PCR and multiplex SBE of provided bisulfite-converted DNA samples; part 3 included the bisulfite conversion step; part 4 required the entire workflow. The corresponding samples for parts 1–4 were produced and sent to participants by the project coordinator. The body fluid type and donor's age for all samples from part 1 and 2, as well as certain samples from part 3, were disclosed to participants to help guide correct data interpretation; however, no information was given for the remaining samples. Consequently, body fluid identification and age prediction were performed consecutively to ensure the right age prediction assay would be selected based on the previously determined body fluid typing results.

Semen, blood and saliva samples were collected from healthy volunteers aged 19–73 years with informed consent according to the method approved by the Institution Review Board of Seoul National University Hospital (IRB no. 1809-074-973 & C-1912-053-1087) and Severance Hospital (IRB no. 4–2015–0083) in Seoul, Korea. Blood samples were obtained by venipuncture and freshly ejaculated semen samples were collected in plastic cups. Aliquots of 1.5 ml blood and semen samples were stored at − 80 °C. Saliva samples were obtained using Oragene™ DNA self-collection kits (DNA Genotek Inc., Ottawa, Canada) and stored at room temperature. Blood and semen swabs were prepared by aspirating 120 μl of body fluid solution onto a sterile cotton-tipped applicator (Puritan Medical Products Co., Guilford, ME), and saliva swabs prepared by aspirating 200 μl of Oragene™ buffer-containing saliva solution onto a sterile cotton-tipped applicator. Body fluid swabs were dried overnight at room temperature on a clean bench and stored at − 80 °C.

For each participating laboratory, a set of 24 test samples was

prepared. To facilitate validation of age prediction, samples with high age prediction accuracy were preferred in sample selection, and samples showing an MAE of 2.5 years in the organizing laboratory were selected. A description of the sample preparation, body fluid type, donor's age, disclosure of body fluid type and age information, and required experiments for each sample is provided in Table 1. Primer mixes sufficient to prepare 100 reactions each for the body fluid identification test and age prediction tests for semen and blood/saliva samples were provided. All samples and primer sets were stored at − 80 °C before being shipped on dry ice to each laboratory.

2.2. DNA extraction and bisulfite conversion

In the organizing laboratory, gDNA was extracted using the QIAamp® DNA Mini kit (Qiagen, Hilden, Germany) and quantified using the Quantifiler™ Human DNA Quantification Kit (Applied Biosystems, Foster City, CA, USA) following manufacturer's instructions. Two hundred ng of gDNA were bisulfite-converted using the EpiTect® Fast DNA Bisulfite Kit (Qiagen, Hilden, Germany) and eluted into 20 µl of Elution buffer. To ensure equal quality and quantity of the samples, gDNA and bisulfite-converted DNA from the same samples were pooled and subsequently divided into aliquots. For each laboratory, gDNA samples were provided at a concentration of 20 ng/µl in a 10 µl volume, and bisulfite converted DNA samples were provided at a concentration of 10 ng/µl in a 10 µl volume. However, the selection of appropriate reagents for DNA extraction, DNA quantification, and bisulfite conversion for the relevant parts of the exercise was left to each laboratory (Supplementary Table S1).

2.3. PCR and SNaPshot multiplex

Example protocols for body fluid identification and age prediction using a SNaPshot multiplex were provided as a reference (Supplementary File S1). The SNaPshot multiplexes for age prediction used were the same as described previously [16,23], but the SNaPshot multiplex for body fluid identification was used with a slight modification of the 7 base deletion at the 5′ end of the reverse MB primer compared to the previous publication [43]. However, laboratories were allowed to choose their own reagents, such as buffers and enzymes.

2.4. Capillary electrophoresis

In the organizing laboratory (Lab 1), SNaPshot products were run on two CE analyzers, the ABI 3130 genetic analyzer (3130 analyzer) and the ABI 3500 genetic analyzer (3500 analyzer) as described in the example protocols (Supplementary File S1). Data were collected using GeneMapper ID v3.2 on the 3130 analyzer and GeneMapper Software 5 on the 3500 analyzer. The results of lab 1 were assigned to 1-1 for data obtained using the 3130 analyzer and 1-2 for data obtained using the 3500 analyzer. Participating laboratories analyzed samples using either the 3130/3130XL (named "3130 laboratories") or the 3500/3500XL (named "3500 laboratories") genetic analyzers under their own optimized conditions, e.g., injection and collection times, as outlined in Supplementary Table S1. Seven laboratories (lab 2, 3, 4, 8, 9, 11 and 12) used the 3130 analyzer, and four laboratories (lab 5, 6, 7 and 10) used the 3500 analyzer.

2.5. Data interpretation

Data interpretation for body fluid identification and age prediction was divided into two steps; the calculation of DNA methylation percentage values followed by body fluid type determination and/or age prediction. Each laboratory was requested to fill in the provided Excel files. The DNA methylation percentage was automatically calculated once the observed peak height values at each marker site were put into the Excel calculator. In the same Excel files, detailed experimental conditions including the thermocycler model, genetic analyzer model, chemicals, kits and the running conditions used by laboratories, were reported accordingly, and these data are summarized in Supplementary Table S1.

Table 1
Overview of the samples and experiments required in each part of the collaborative exercise.

Part	No.	Sample	Comment	Fluid type	Donor age	Information	Required experiments
1	1	SE-bf-SBE	SBE product for a reference	Semen	NA[c]	Disclosed	Capillary electrophoresis (CE)
	2	BL-bf-SBE	SBE product for a reference	Blood	NA[c]	Disclosed	
	3	SA-bf-SBE	SBE product for a reference	Saliva	NA[c]	Disclosed	
	4	SE-age-SBE	SBE product for a reference	Semen	52	Disclosed	
	5	BL-age-SBE	SBE product for a reference	Blood	23	Disclosed	
	6	SA-age-SBE	SBE product for a reference	Saliva	40	Disclosed	
2[a]	7	SE-young-cD10	bisulfite-converted DNA	Semen	47	Disclosed	Multiplex PCR, Multiplex SBE, CE
	8	SE-old-cD10	bisulfite-converted DNA	Semen	52	Disclosed	
	9	BL-young-cD10	bisulfite-converted DNA	Blood	23	Disclosed	
	10	BL-old-cD10	bisulfite-converted DNA	Blood	56	Disclosed	
	11	SA-young-cD10	bisulfite-converted DNA	Saliva	24	Disclosed	
	12	SA-old-cD10	bisulfite-converted DNA	Saliva	60	Disclosed	
3[b]	13	SE-young-gD	genomic DNA	Semen	33	Disclosed	Bisulfite conversion,
	14	SE-old-gD	genomic DNA	Semen	63	Disclosed	Multiplex PCR,
	15	BL-young-gD	genomic DNA	Blood	28	Disclosed	Multiplex SBE,
	16	BL-old-gD	genomic DNA	Blood	53	Disclosed	CE
	17	SA-young-gD	genomic DNA	Saliva	40	Disclosed	
	18	SA-old-gD	genomic DNA	Saliva	60	Disclosed	
	19	gD-TEST01	genomic DNA	Blood	39	Undisclosed	
	20	gD-TEST02	genomic DNA	Saliva	58	Undisclosed	
	21	gD-TEST03	genomic DNA	Semen	41	Undisclosed	
4	22	swab-TEST01	body fluid swab	Saliva	30	Undisclosed	DNA extraction, Bisulfite conversion, Multiplex PCR, Multiplex SBE,
	23	swab-TEST02	body fluid swab	Blood	33	Undisclosed	CE
	24	swab-TEST03	body fluid swab	Semen	44	Undisclosed	

[a] Samples were provided at a concentration of 10 ng/µl in a 10 µl volume.
[b] Samples were provided at a concentration of 20 ng/µl in a 10 µl volume.
[c] NA, not applicable.

2.5.1. Body fluid identification

Body fluid types were determined according to each laboratory's interpretation by confirming the presence of methylation signals at the target body fluid marker sites. In the organizing laboratory, a detection threshold for methylation and non-methylation signals was set at 100 RFU (Relative Fluorescence Units) for both 3130 and 3500 genetic analyzers, and an interpretational threshold for a positive methylation signal was set at 10%, based on conclusions from the previous collaborative exercise [42]. Thereafter, the type of body fluid was determined when the results were positive at two target marker loci for cases involving semen and blood, and at one target marker locus for cases involving saliva.

2.5.2. Age prediction

From peak height values put into the Excel calculator, ages were predicted automatically using the age predictive algorithms implemented in the Excel worksheet. Information on the application of these algorithms and CpG markers can be found in Supplementary File S1. The semen age algorithm has been published [23], as well as those for blood and saliva [16,44]. Unlike body fluid identification, no specific peak height threshold was applied to detect methylated and non-methylated peaks for age estimation. However, peak heights too low to be measured were counted as zero.

2.5.3. Statistical analysis

For statistical analysis of the methylation values obtained at each marker, one-way analysis of variance (ANOVA) with *post-hoc* Tukey HSD test was conducted using the data which followed normality. If the methylation values did not follow normality, the Friedman test was performed to confirm whether the mean methylation values among laboratories were statistically significantly different. All statistical tests were conducted using IBM SPSS Statistics 26 and Python 3.9.8, with the Python packages used comprising pandas, matplotlib, scipy and statsmodels.

3. Results and discussion

3.1. Body fluid identification

Each of the twelve participating laboratories successfully completed the required steps for all four parts. However, differences in the results occurred because no specific guideline was provided for the thresholds to detect positive peaks and methylation signals, which led to laboratories setting different detection thresholds between 50 and 100 RFU, and some laboratories recorded a positive methylation signal even when a marker had a methylation percentage of less than 10%. In addition, certain laboratories doubled or quadrupled the recommended amounts of bisulfite-converted DNA (10 ng in 1 μl), PCR product (1 μl) or SBE product (1 μl) in their experiments.

The body fluid typing results reported from each laboratory are collated in Table 2, Supplementary Figs. S1–S3 and Supplementary Table S2. With or without the provision of body fluid type information (see Table 1), all laboratories produced an accurate determination of body fluid type, the information necessary to select the correct subsequent age prediction assay, in almost all samples. Several marker loci exhibited low methylation signals even in non-corresponding body fluids. However, the methylation percentages were mostly less than 10% and thus, easy to assign as artifacts. Some artifact peaks were also detected with higher methylation levels, but these peaks did not interfere with the correct determination of body fluid types because the type of body fluid was determined only when the results were positive at two target marker loci for cases involving semen and blood. (Supplementary Figs. S1–S3).

3.1.1. Part 1: capillary electrophoresis (CE)

The aim of analyzing prefabricated SBE products in part 1 of the exercise was to use this assay to optimize the CE running conditions for the genetic analyzer used, and to let the obtained CE results serve as a reference for the interpretation of results from the subsequent parts. The results from all laboratories were highly comparable. Eleven laboratories demonstrated similar methylation percentages at all marker sites and produced 10% or greater methylation percentage values at all target

Table 2
The collated body fluid typing results from participating laboratories. Green, yellow, and red boxes represent correct, inconclusive, and failed typing results, respectively.

Part	No.	Sample	Lab no.												
			1-1	1-2	2	3	4	5	6	7	8	9	10	11	12
1	1	SE-bf-SBE													
	2	BL-bf-SBE													
	3	SA-bf-SBE													
2	7	SE-young-cD10													
	8	SE-old-cD10													
	9	BL-young-cD10													
	10	BL-old-cD10													
	11	SA-young-cD10													
	12	SA-old-cD10													
3	13	SE-young-gD													
	14	SE-old-gD													
	15	BL-young-gD													
	16	BL-old-gD													
	17	SA-young-gD													
	18	SA-old-gD													
	19	gD-TEST01													
	20	gD-TEST02													
	21	gD-TEST03													
4	22	swab-TEST01													
	23	swab-TEST02													
	24	swab-TEST03													

marker sites for the corresponding body fluids. However, lab 12 detected an unexpectedly high methylation percentage at the BL1 marker due to the presence of a non-methylation peak below their detection threshold. Although all laboratories demonstrated similar methylation percentages at target marker sites with less than ± 7% methylation standard deviation, the relative methylation percentages from 3130 analyzers were higher than those from 3500 analyzers. This phenomenon is due to the differences in inter-color balance, which altered the peak height ratio between dyes for each genetic analyzer type. Such differences have been reported to occur due to variable binning [44,45], a process which allows more pixels of light to be collected from the CCD camera in weak color channels, thereby reducing signal variation and helping balance dyes with different fluorescent efficiencies [44,46]. Therefore, the CE results of this study are presented separately for the two genetic analyzers used.

3.1.2. Part 2: applying the Single Base Extension (SBE) reaction

In this part, six bisulfite-converted DNA samples with information on body fluid type and age were provided and both the SBE reaction and subsequent CE were performed. Eleven laboratories reported the correct body fluid typing results, and lab 7 reported "inconclusive" data for the BL-old-cD10 sample due to failure of the SBE reaction (Table 2). We indicated the body fluid typing result of this sample from lab 7 as failed in Table 2, because only one locus showed a detectable peak. This failure seems to be the result of carry-over of inhibitors from the bisulfite conversion, as lab 7 used 4 µl of converted DNA in 10 µl PCR volume instead of the recommended 1–4 µl converted DNA in 20 µl PCR volume. Such inhibition effect has been shown for some kits, e.g., the Sigma Imprint DNA modification kit and the EpiTect® Fast DNA Bisulfite Kit, which was used in preparation of the part 2 samples (see the 2019 ISFG workshop materials outlined in Supplementary Figs. S4–S5). In contrast, lab 2 observed either none or only one of the methylation and non-methylation peaks at the BL3 marker in BL-old-cD10 and BL-young-cD10 samples, thereby reporting none and 100% methylation, respectively. As with lab 12 in the part 1, the lab 2 results for blood samples were attributed to low SBE signals at the BL3 marker.

In the saliva sample results, lab 4 using another PCR kit (QIAGEN Multiplex PCR plus Kit), reported correct body fluid typing results, but had a significantly lower methylation percentage (p-value < 0.05) at the SA1 marker than the other 3130 laboratories (Supplementary Tables S3 and S4, Supplementary Fig. S9). This lower methylation percentage was maintained in all experiments which involved the PCR and SBE analyses. According to the information provided by Qiagen (https://www.qiagen.com/us/products/discovery-and-translational-research/pcr-qpcr-dpcr/pcr-enzymes-and-kits/end-point-pcr/qiagen-multiplex-pcr-plus-kit/), the multiplex PCR buffer of the QIAGEN Multiplex PCR plus Kit contains an optimized combination of potassium and ammonium ions (K^+ and $NH4^+$). Since ammonium ions can destabilize weak hydrogen bonds between mismatched base pairs during the PCR annealing step, this buffer composition minimizes non-specific annealing and stabilizes specifically bound primers, thereby ensuring a high ratio of specific to non-specific primer binding. However, due to the presence of CpGs in the primer binding regions, the forward and reverse primers of the SE2, VF1, SA1, and MB markers include R (A/G) or Y (C/T) nucleotides in their sequences. Primer degeneracy incorporated in the designs was highest in VF1 (8 nucleotide sequence combinations), followed by SA1 (4), and SE2 (2)/MB (2), and the pool of PCR amplicons obtained from these degenerate primers naturally include amplicons both from specific and non-specific primer annealing. In contrast, the PCR products generated using the QIAGEN Multiplex PCR plus Kit may have produced lower levels of non-specific primer annealing, resulting in the different DNA methylation percentages observed in lab 4 at the SA1 marker.

In general, the laboratories with a detection threshold of 50 RFU (labs 2, 3, 9, 11, and 12) showed unexpected methylation signals more frequently, regardless of the type of genetic analyzers. Since all unexpected methylation signals could be clearly determined as artifacts as explained above, all samples were correctly interpreted. However, the number of unexpected methylation signals detected in saliva samples was more than twice the detected unexpected signals in semen or blood samples. These were mostly caused by unexpected methylation peaks for VF1 and VF2, which accounted for the largest portion of unexpected methylation signals observed in saliva. These unexpected methylation peaks tend to be observed from part 2, involving the PCR and SBE reactions, and therefore, may be considered to be the result of stochastic amplification of background methylation that is very low but inherently present at non-target marker loci of a particular body fluid.

Another phenomenon found in the results from the part 2 results of lab 12 was detection of many non-specific peaks around the region including SE1 and SE2 markers. This observation, involving the presence of non-specific SBE peaks, may be the result of carry-over of PCR primers left intact by the incomplete purification of the PCR products.

3.1.3. Part 3: bisulfite conversion

In the part 3, six gDNA samples with known body fluid types and three gDNA samples with undisclosed body fluid types were analyzed. Unlike part 2, the bisulfite conversions were completed by each laboratory. Due to the increase of experimental steps required, the measured DNA methylation percentages varied more (Supplementary Table S2), but 98% of the fluids (104 of 108 cases) were correctly typed (Table 2). Only three laboratories reported inconclusive body fluid typing results; lab 4 reported "unsure, but blood or saliva" for gD-TEST02 (saliva) because BL1 showed a higher methylation percentage than SA1 (19% vs. 16%) with non-methylation at BL3; lab 11 reported "saliva-inconclusive" for SA-young-gD, because of a low methylation signal at SA1 (11%) and the presence of additional low methylation signals (< 10%) at BL1, BL3, and VF2 loci. In addition, lab 12, as in the second part, showed too many non-specific SBE peaks in the region that included SE1 and SE2 markers, therefore difficulty was found in specifying the peaks at the target marker sites. When these phenomena occurred in semen samples, i.e., SE-young-gD and SE-old-gD, accurate body fluid typing was not feasible. For lab 7 which used varying volumes of bisulfite-converted DNA, relatively higher or lower methylation peaks were observed at target marker sites compared to other laboratories.

As in the previous parts, the presence of unexpected methylated peaks, in practice, did not adversely affect correct body fluid typing in most samples, but 10% or greater methylation signals at both VF1 and VF2 loci were observed in BL-old-gD from labs 3 and 9, and in gD-Test01 (saliva) from lab 9. Additional unexpected peaks were most frequently observed at the BL1 and VF1 loci (23% and 32% of the total non-specific peaks in this part). However, the repeated observation of additional unexpected peaks at specific marker sites in the same laboratory, e.g., BL1 marker site in lab 9, were considered the result of incomplete bisulfite conversion or contamination.

Among the twelve laboratories, six laboratories (1, 4, 5, 9, 10, and 11) used a bisulfite conversion kit from Qiagen, four laboratories (2, 6, 7, and 12) used a kit from Zymo Research, and two (3 and 8) used a kit from Promega. There appears to be no significant difference in the results between laboratories using different bisulfite conversion kits (p-value > 0.05) (Supplementary Tables S3 and S5, Supplementary Fig. S10). However, labs 3 and 8 with the Promega kits sometimes showed lower methylation in SA1, BL1 and BL3 markers in comparison with other laboratories.

3.1.4. Part 4: the whole workflow

For part 4, three swabs of unknown body fluid types had to be investigated. Ten laboratories correctly identified the body fluid type of each swab, and two laboratories reported the results that could be interpreted to be inconclusive; lab 7 did not observe any peaks at the BL3 locus in all three swabs, and lab 12 did not observe any peaks at the SE2 locus in swab-TEST01 and swab-TEST03.

However, it was encouraging that the laboratories that used different reagents and genetic analyzers all generated successful body fluid typing

results. Lab 10 used three different kits for DNA extraction (PrepFiler™, DirEx™, and MinElute PCR purification kit): one for each of the three swab samples. Even when using the DirEx™ kit (GeneAll, Seoul, Korea), which allows direct PCR or bisulfite conversion reaction without extraction and quantification of gDNA for swab-TEST02, this laboratory still successfully obtained a body fluid typing result. In addition, lab 11 performed body fluid identification using DNA extracted from two fractions of a differential extraction of swab-TEST03. Since the body fluid of swab-TEST03 only had semen present, both fractions, the precipitate and supernatant, produced the same result.

3.2. Age prediction

All participating laboratories successfully performed subsequent age prediction experiments using the results obtained from the body fluid type information obtained in previous experiments. As with the body fluid typing experiments, there was a noticeable difference in the measured values between 3130 and 3500 analyzers (Fig. 1). Therefore, methylation percentages are provided separately for the two genetic analyzer types (semen), or the methylation percentages were adjusted from the 3500 analyzers to 3130 analyzers (blood and saliva) by applying regression equations to correct the methylation values [44] (Figs. 1 and 2). Among the body fluid types, the markers for semen age prediction performed best, with no measurement failure in any of the laboratories.

3.2.1. Part 1: capillary electrophoresis (CE)

As in the body fluid identification experiments, CE analysis of the prefabricated SBE products was optimized in order to obtain best conditions for the subsequent experiments. As expected, the results from this part displayed the lowest differences between laboratories (Figs. 1 and 2). All laboratories using the same genetic analyzer reported similar CE analysis results, despite using different running conditions (Supplementary Table S1). The standard deviations of the methylation percentages among all laboratories using the same genetic analyzer were 5% or less in all assays of this part (Fig. 1). However, lab 12 failed to produce CE result for BL-age-SBE sample and showed very unusual DNA methylation values in parts 2–4. Therefore, data for this lab was presented, but excluded from the calculation of predicted ages for blood and saliva samples in parts 1–4. Lab 2 applied a detection threshold of 50 RFU, and initially did not include a methylated peak in the blue channel at the *MIR29B2CHG* of BL-age-SBE (40 RFU). However, when the results were merged in the organizing laboratory, the peak was determined as correct and used for age prediction. Lab 7 reported zero percent methylation at *KLF14* locus in BL-age-SBE and SA-age-SBE samples, and lab 9 reported slightly lower methylation values at all observed marker sites compared to the other 3130 laboratories (Fig. 1).

It is noteworthy that differences in the measurement levels of the DNA methylation can affect the age prediction. This is because the prediction models used in this study are based on reference data generated only with a 3130 analyzer [16,23]. In practice, when applying the 3500 analyzer data to a 3130 analyzer age prediction model, the MAEs of 3500 laboratories for blood and saliva samples were very large. As the result, the average predicted ages of 3500 laboratories were much lower than those of 3130 laboratories; differences between genetic analyzers were 6.6 years for blood and 7.4 years for saliva. However, in the case of semen, the difference of average predicted ages between the two genetic analyzers was below 3 years and the standard deviation of all laboratories was as small as 1.6 years. The semen model consists of three markers, *TTC7B* and cg12837463 having a negative association with age, and *NOX4/FOLH1B* having a positive association with age. In addition, the positive correlation coefficient value and negative correlation coefficient values in the prediction equation are similar. Therefore, it seems likely that the positive and negative associations with methylation value offset the observed methylation changes each other in spite of which of genetic analyzer used. For age prediction in blood and

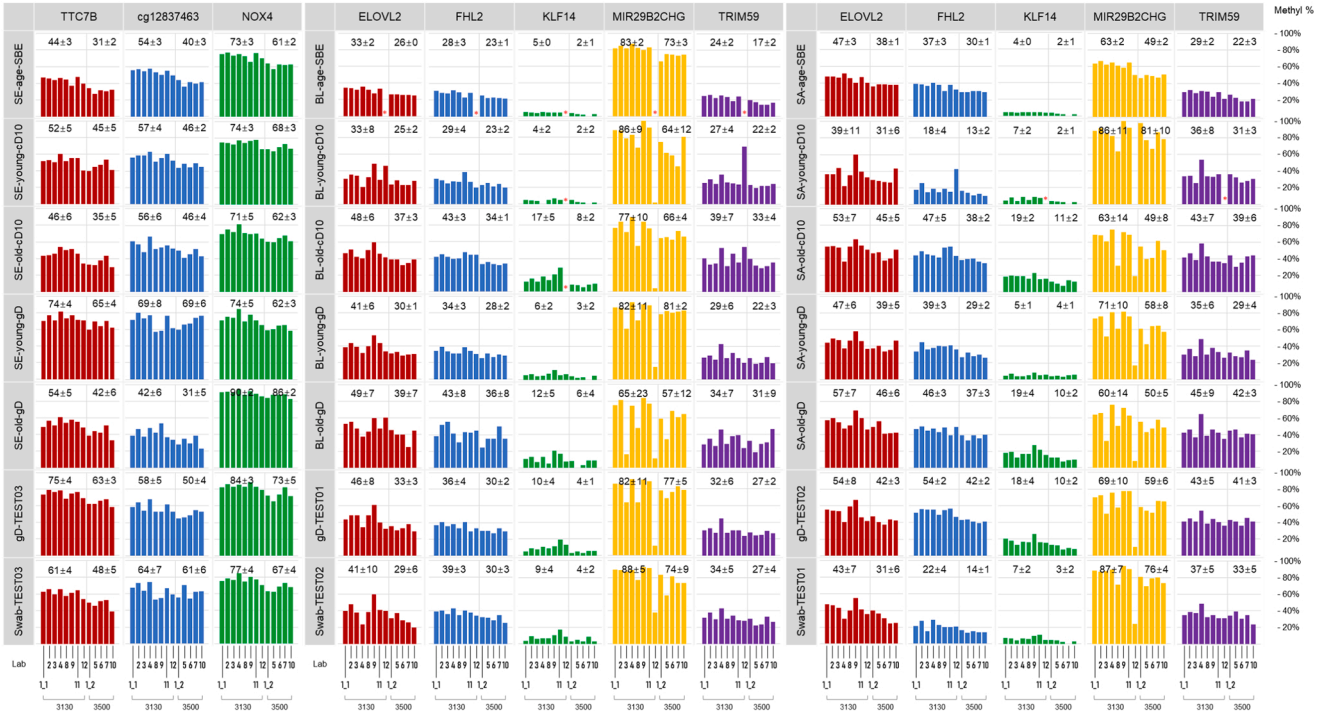


Fig. 1. DNA methylation percentages of semen (left), blood (middle) and saliva (right) age prediction from each participating laboratory. The data were presented separately for the two genetic analyzer types used (3130 and 3500 genetic analyzers). The x-axis shows laboratories and the y-axis the methylation percentages observed. The methylation percentages that could not be calculated due to peak drop-out were put as 0 and marked with red asterisks. Average methylation values and standard deviations are also indicated. However, due to the failure in CE settings, the lab 12 data for blood and saliva samples were not included in the calculation.

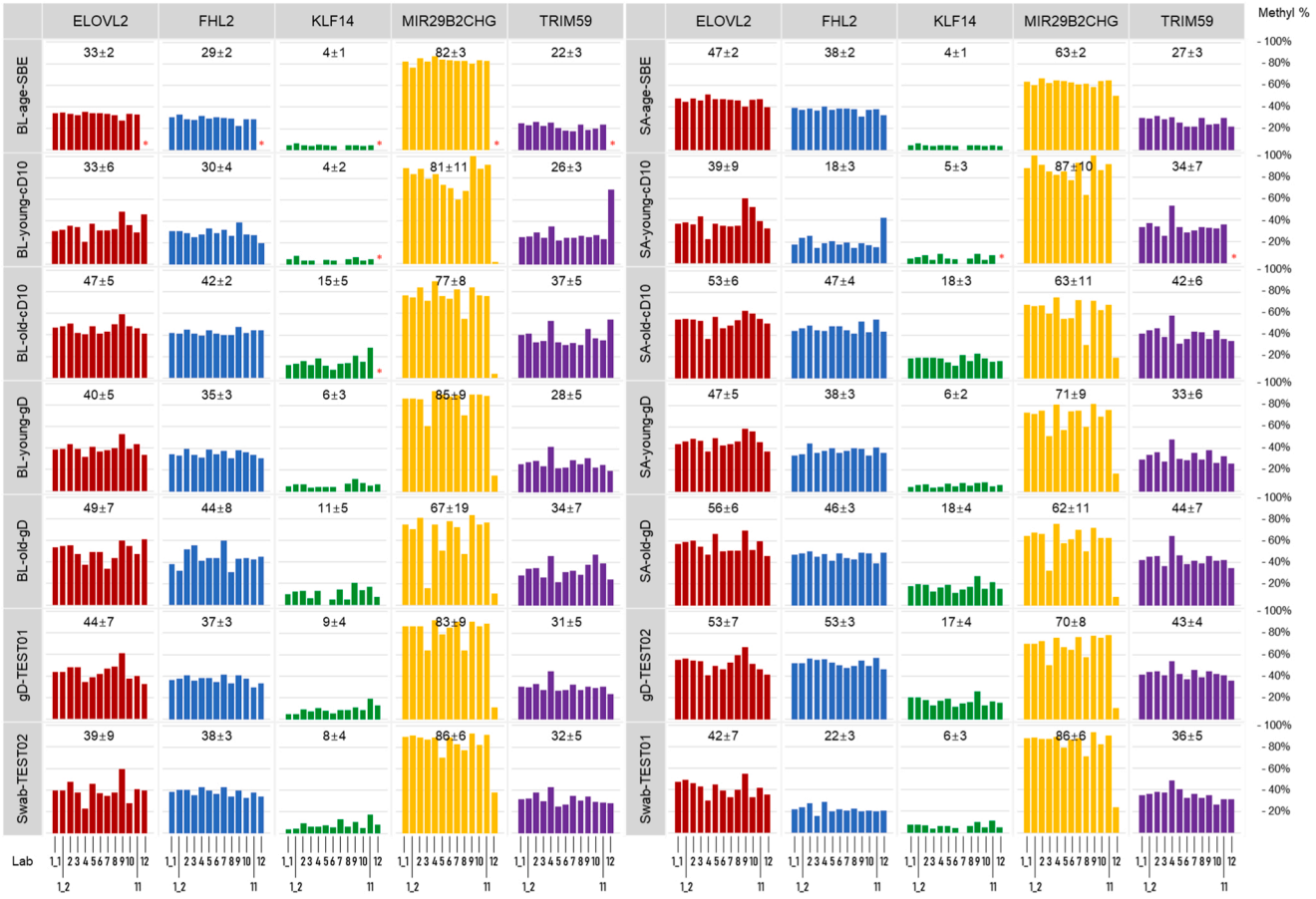


Fig. 2. Reanalyzed DNA methylation percentages of blood (left) and saliva (right) age prediction from each participating laboratory. Because the data of 3500 genetic analyzers were adjusted to correlate with the 3130 genetic analyzers, the data were not separated for 3130 or 3500 laboratories. The x-axis shows laboratories and the y-axis the methylation percentages observed. The methylation percentages that could not be calculated due to peak drop-out were put as 0 and marked with red asterisks. Average methylation values and standard deviations are also indicated. However, due to the failure in CE settings, the lab 12 data were not included in the calculation.

saliva samples, equations to adjust the methylation values from 3500 analyzer data to 3130 analyzer data are available [44]. Using this method resulted in the average predicted age difference between the two analyzers in both blood and saliva being lowered by approximately 2 years.

3.2.2. *Part 2: the Single Base Extension (SBE) reaction*

In part 2, two bisulfite-converted DNA samples with age information (47 years and 52 years) were provided for age prediction from semen. The age prediction results were highly accurate and similar in all laboratories (Fig. 3). However, three laboratories showed higher variation at a single CpG site in comparison to the other labs, namely labs 7 and 12 at *TTC7B* and lab 8 at cg12837463. Since the laboratories applied the reagents that the other laboratories also used, the differences in these laboratories appeared to be caused by unreported difference in experimental conditions or analyst's proficiency. For these three laboratories, these effects were also seen in parts 3 and 4.

For the age prediction test using blood and saliva, four bisulfite-converted DNA samples including two blood and two saliva samples with age information (23, 56, plus 24 and 60 years, respectively) were provided. In contrast to the first experimental part, problems appeared in several laboratories. Firstly, peak drop-outs at the *KLF14* marker were observed for samples BL-young-cD10 and SA-young-cD10 in labs 4, 7, and 9. In addition, an inter-laboratory comparison using methylation values adjusted from 3500 analyzers to 3130 analyzers showed some laboratories had increased or decreased methylation when compared to others (Fig. 2) (Supplementary Table S6, Supplementary Fig. S11): lab 4 reported significantly lower and higher methylation in *ELOVL2* and *TRIM59* markers, respectively (p-value < 0.05). Lab 9, which showed lower methylation in part 1, had relatively higher methylation in *ELOVL2* and *MIR29B2CHG* markers. As explained above, the results of lab 4 are considered to be due to use of the QIAGEN Multiplex PCR plus Kit. Among five CpG markers, only *ELOVL2* and *TRIM59* markers were amplified using degenerate forward and reverse primers. However, the cause of lab 9 results was not clear and continued in the subsequent experiments. Lab 8 had a significantly lower methylation at *MIR29B2CHG* and showed similar tendencies through parts 2–4 (p-value < 0.05). In practice, such results can occur from the formation of primer dimers during the SBE reaction, which lacks the hot start process. In the SNaPshot multiplex for blood and saliva age prediction, the SBE primers for the *ELOVL2* and *MIR29B2CHG* markers could bind to each other and show an unexpected green peak (associated with non-methylation) at the *MIR29B2CHG* locus, which then results in a much lower methylation value than is actually the case in the sample (Supplementary Fig. S6). Therefore, SBE reactions should be prepared on ice.

Due to the increased variation in DNA methylation between laboratories, standard deviations of predicted ages also increased in part 2 but were still less than 6 years (Fig. 3).

3.2.3. *Part 3: bisulfite conversion*

In part 3, nine genomic DNA samples, comprising six with known body fluid type and age, plus three with undisclosed body fluid type and age were provided. Each laboratory used their own choice of bisulfite conversion kit.

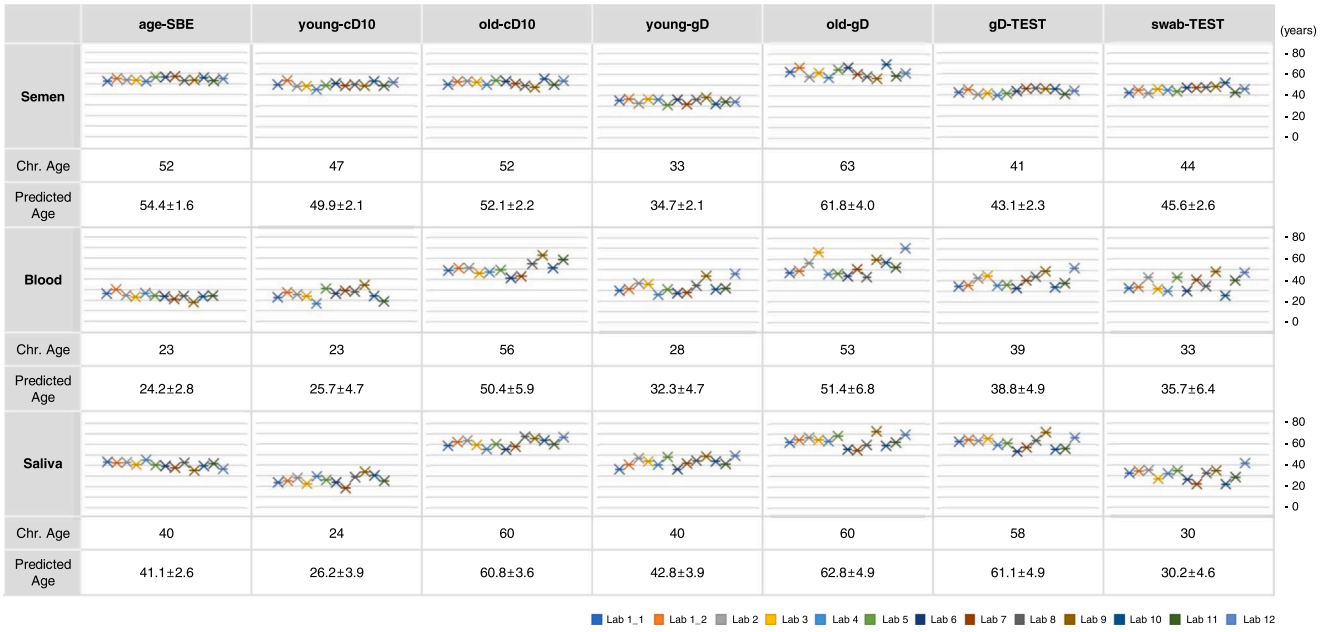


Fig. 3. Age estimation results of semen (top), blood (middle) and saliva (bottom). The x-axis represents the laboratories and the y-axis the predicted ages (in years). The chronological age is abbreviated as "Chr. Age". The mean and standard deviation of predicted ages calculated are indicated for each sample and when the values were calculated, the data of lab 12 was not included for blood and saliva (predicted ages remain in the graph). In case of 3500 laboratories, the methylation values were adjusted to correlate with the 3130 analyzers before age calculation.

In the experiments with semen samples, all laboratories reported accurate age prediction results with an average prediction error below 3 years and a standard deviation among laboratories also below 4 years. Although the experimental procedure was more complex compared to the previous parts, the observed variation in DNA methylation between laboratories using the same genetic analyzer was still small, and the standard deviations of predicted ages were also 3 years or less.

In contrast to the semen samples analyzed, the DNA methylation values measured in the blood and saliva samples displayed more variability between laboratories. Blood DNA samples showed the largest variability in age prediction with a standard deviation of 5.5 years (Fig. 3). This was mainly caused by highly diverging methylation measurements at the *MIR29B2CHG* locus in several labs, and unusually low methylation was noticeable in the results from labs 3 and 8 for all six blood and saliva samples. These two laboratories used the MethylEdge® Bisulfite Conversion System from Promega, but lab 8 had also shown such a tendency in part 2. Therefore, it was not clear whether the low methylation levels recorded in labs 3 and 8 was attributable to the use of a particular bisulfite conversion kit.

All laboratories correctly predicted the gD-TEST01 sample as blood and gD-TEST02 sample as saliva, and all applied the right SNaPshot multiplex for age prediction, with resulting MAEs below 5 years for both samples.

3.2.4. Part 4: DNA extraction

In part 4, three body fluid swab samples with undisclosed origin and age information were provided, and the age prediction test was performed based on the body fluid identification results. The swab-TEST01 and the swab-TEST02 were correctly identified to be from saliva and blood, respectively. MAEs were 4.2 years for saliva and 5.5 years for blood. The swab-TEST03 sample was correctly identified as semen, and the predicted ages from all twelve laboratories were close to the chronological age with an MAE of 2.5 years. Because MAE and standard deviation values did not increase in comparison to the previous parts, it was concluded that there was no noticeable difference arising from the process of DNA extraction and quantification applied by each laboratory.

4. Conclusion

The results of body fluid identification from each laboratory were mostly conclusive (98%) and provided sufficient information to determine the appropriate age prediction methods to choose for subsequent analyses. Non-specific peaks occurred in some samples from certain laboratories, but those peaks did not affect the analysis and interpretation of body fluid typing results. Rather, the input volume of bisulfite-converted DNA used for PCR amplification appears to affect the success of the analysis, and the PCR buffer composition affects the measurement of DNA methylation. Unlike body fluid identification, which simply confirms the presence of positive methylation signals at target marker sites, age prediction requires quantitative measurements of DNA methylation and therefore showed greater differences between laboratories. In CE analysis a systematic measurement difference related to the type of genetic analyzer used was identified as the largest source of variation between laboratories. This difference particularly affected the age prediction results of blood and saliva samples, and therefore, DNA methylation values produced by 3500 analyzers were adjusted to correlate with 3130 analyzer-based data at five blood and saliva age-associated CpGs using appropriate equations.

DNA methylation variation between laboratories was evident from the experimental parts involving PCR and SBE analysis. As in body fluid identification analysis, using the buffer that ensures a high ratio of specific to non-specific primer binding resulted in the observed changes in DNA methylation measurement, especially in the PCR step using degenerate primers, e.g., *ELOVL2* and *TRIM59*. In addition, high input volumes of bisulfite-converted DNA often caused PCR failure presumably due to the carry-over of PCR inhibitors from bisulfite conversion reaction. The proficiency of the analysts and the experimental conditions for SNaPshot reaction were also found to be important for consistent DNA methylation measurement, and the methylation percentage data at *MIR29B2CHG* locus was most affected by the experimental conditions. Several bisulfite conversion kits were used for this study, but the difference produced by the use of a specific kit was not clear.

Even when different experimental settings were used in each laboratory, it was encouraging that the overall error across all four parts was

only 2.7 years for semen, 5.0 years for blood and 3.8 years for saliva (Supplementary Figs. S7 and S8). This technical validation study has enabled us to detect many of the potential factors that influence methylation measurements in each experimental part, indicating that such variation can be better controlled by harmonizing appropriate reagents and experimental conditions, as well as improved technical training.

Conflict of interest

The authors have declared no conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.fsigen.2021.102656.

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